



TeamUp
Consultants



CYBERSECURITY CAREER PLAYBOOK

A PRACTICAL ROADMAP FROM BEGINNER
TO JOB-READY CYBERSECURITY PROFESSIONAL



START YOUR JOURNEY

Build a strong foundation and understand the cybersecurity landscape.



DEVELOP IN-DEMAND SKILLS

Learn essential tools, techniques, and real-world practices across multiple domains.



BUILD REAL EXPERIENCE

Work on practical projects, build a home lab, and create a portfolio that stands out.



LAND YOUR DREAM ROLE

Prepare your resume, ace interviews, and take the next step in your cybersecurity career.



FOUNDATIONS

Build strong basics



PRACTICAL SKILLS

Hands-on tools & techniques



REAL PROJECTS

Build, document & showcase your work



CAREER GROWTH

Get hired & elevate your future

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01 - Why Cybersecurity, Why Now

What Is Cybersecurity?

Cybersecurity is the practice of protecting digital systems, networks, and data from unauthorized access, theft, and damage. It encompasses technology, processes, and people working together to defend against threats ranging from individual hackers to organized criminal groups and nation-states. In today's connected world, cybersecurity is not optional - it is essential infrastructure.

Who This Playbook Is For

This playbook is designed for **career changers, recent graduates, and IT professionals** who want to transition into cybersecurity. You do not need prior security experience. You need curiosity, persistence, and willingness to learn hands-on technical skills. Commit 10–15 hours per week for 12 months and you can reach job-ready status.

How to Use This Playbook

Read sequentially - each section builds on the previous one. Customize the pace to your situation. If you already know networking, accelerate through that section. If you're new to Linux, spend extra time there. The goal is not speed - it is **depth and practical competence**.

Common Beginner Misconceptions

"I need a CS degree."

Many security professionals come from non-technical backgrounds. Certifications and practical projects matter more than degrees.

"I need to be a hacker first."

Most cybersecurity careers are defensive. You learn security concepts, not hacking for its own sake.

"Certs guarantee a job."

Certifications open doors, but practical projects and communication skills close the deal.

"I need to know everything first."

Entry-level roles expect learning on the job. You need foundational knowledge and demonstrated willingness to grow.

What You Will Achieve by the End of This Playbook

1

Master Foundations

Computer fundamentals, networking, Linux, and Windows

2

Build a Home Lab

Functional security lab with real tools and projects

3

Create a Portfolio

Professional GitHub, resume, and LinkedIn presence

4

Land the Role

Pass technical and behavioral interviews for entry-level security positions

02 - The Cybersecurity Career Landscape

Cybersecurity is not one job - it is a diverse field with **15+ distinct career paths**. Understanding these paths helps you choose the right specialization early. Entry-level roles exist in SOC, Cloud Security, GRC, and Vulnerability Assessment. Offensive roles typically require 2-3 years of experience.

Defensive Security - Blue Team

SOC Analyst

Monitors networks for threats, triages alerts, responds to incidents. **Entry-level.** Requires strong attention to detail and ability to work under pressure. Salary: **\$55K-\$75K**

Incident Response Specialist

Investigates security breaches, contains damage, recovers systems. Requires deep technical knowledge and calm decision-making. Salary: **\$80K-\$120K**

Threat Detection Engineer

Builds detection rules and signatures to identify attacks. Requires SIEM expertise and understanding of attacker behavior. Salary: **\$85K-\$130K**

Security Engineer

Designs and implements security infrastructure. Requires networking knowledge and system administration experience. Salary: **\$90K-\$140K**

Offensive Security - Red Team

Penetration Tester

Authorized to attack systems and find vulnerabilities before attackers do. Requires deep technical skills and business communication. Salary: **\$90K-\$150K**

Vulnerability Assessor

Scans systems for weaknesses and reports findings. **Entry-level offensive role.** Good stepping stone to penetration testing. Salary: **\$65K-\$95K**

Red Team Operator

Simulates advanced attackers in full-scope engagements. Requires extensive experience and creative problem-solving. Salary: **\$120K-\$180K**

AppSec / DevSecOps / GRC

Specialized domains covering application security, pipeline security, and compliance. Each requires different skill combinations. Salary: **\$60K-\$150K+**

Career Path Comparison at a Glance

Career Path	Entry Level?	Key Skills	Salary Range	Track
SOC Analyst	✔ Yes	alert triage, SIEM, networking	\$55K-\$75K	Blue Team
Incident Response	⚠ No	forensics, Linux, networking	\$80K-\$120K	Blue Team
Penetration Tester	⚠ No	networking, Linux, exploitation	\$90K-\$150K	Red Team
Cloud Security	✔ Yes	AWS/Azure, IAM, networking	\$85K-\$130K	Cloud
AppSec Engineer	⚠ No	coding, OWASP, secure SDLC	\$95K-\$150K	AppSec
GRC Analyst	✔ Yes	risk, compliance, documentation	\$60K-\$90K	GRC

03 - Self-Assessment, Career Selection & Cybersecurity Foundations

Career Path Self-Assessment

Before committing to a specialization, assess your natural strengths honestly. A successful SOC analyst may struggle as a penetration tester, and vice versa.

Do you enjoy troubleshooting systems?

→ SOC Analyst, Incident Response, Security Engineering

Do you prefer attacking or defending?

→ Attacking: Red Team, Pen Testing.
Defending: SOC, IR, Security Engineering

Are you interested in code and development?

→ AppSec, DevSecOps, Vulnerability Assessment

Do you prefer business-focused work?

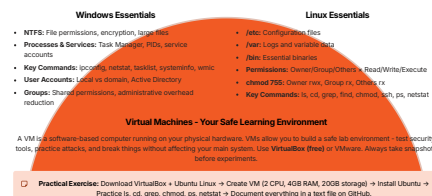
→ GRC Analyst, Risk Manager, Compliance Officer

Do you enjoy research and investigation?

→ Threat Intelligence, Forensics, Incident Response

Phase 1: Computer Foundations - Why They Matter

Before learning security, you must understand how computers work. Security is built on systems administration, networking, and operating systems knowledge. Skipping foundations leads to confusion later.



Foundations Completion Checklist

Hardware CPU, RAM, storage, NIC fundamentals understood	Windows File system, permissions, processes, CLI mastered
Linux File system, permissions, 13 core commands mastered	Virtualization Hypervisor installed, Linux VM running, snapshots configured

04 - Networking, Security Fundamentals & Essential Tools

The OSI Model - 7 Layers Every Security Pro Must Know

#	Layer	Function	Examples
7	Application	User applications	HTTP, HTTPS, DNS, SSH
6	Presentation	Data formatting, encryption	SSL/TLS
5	Session	Session management	Tokens, cookies
4	Transport	End-to-end communication	TCP, UDP, ports
3	Network	Routing, IP addressing	IP, ICMP, routing
2	Data Link	MAC addressing, switching	MAC, ARP, switches
1	Physical	Physical transmission	Cables, hubs, signals

TCP/IP Addressing

- **IPv4:** 32-bit address, four octets (e.g. 192.168.1.100)
- **CIDR:** /24 = 255.255.255.0, /16 = 255.255.0.0, /8 = 255.0.0.0
- **Private Ranges:** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
- **IPv6:** 128-bit hex address - more addresses, improved security

The CIA Triad - Core Security Principles

Confidentiality

Only authorized people access data. Encryption + access control.

Integrity

Data is accurate and unmodified. Hashing + digital signatures.

Availability

Systems accessible when needed. Redundancy + disaster recovery.

Essential Security Tools

Nmap - Network Scanner

Discovers hosts, open ports, and services. Beginner use: scan your home network. Core commands: `nmap target.com`, `nmap -sV target.com`. **Only scan networks you own.**

Wireshark - Packet Analyzer

Captures and analyzes network traffic at packet level. Filter with `tcp.port == 80`. Beginner use: analyze HTTP and DNS on your own network.

Burp Suite - Web Security Testing

Intercepts and analyzes HTTP traffic. Industry standard for web security testing. Practice with OWASP WebGoat or Juice Shop.

SIEM - Splunk / ELK Stack

Collects, normalizes, and correlates logs across an organization. ELK Stack is free and ideal for learning. Learn: log collection, correlation rules, dashboards, alerting.

Git + GitHub

Version control and portfolio hosting. Create repositories for every project. Write detailed README files. Employers expect GitHub proficiency.

Docker

Containerization for modern security labs. Run tools in isolated containers. Essential for DevSecOps track. Learn: pulling images, running containers, Dockerfiles.

Common Ports to Memorize

Port	Protocol	Purpose
22	SSH	Secure remote access
80	HTTP	Unencrypted web traffic
443	HTTPS	Encrypted web traffic
3389	RDP	Remote Desktop Protocol
3306	MySQL	Database access

📌 **Critical Ethical Reminder:** Security testing must only be performed against systems you own or have explicit written authorization to test. Unauthorized testing is illegal and unethical.

05 - Cybersecurity Specialization Tracks

You now have foundational knowledge. It is time to specialize. Choose **one primary track** and build expertise. Depth in one area is more valuable than shallow knowledge in many. You can explore other tracks later.



Blue Team - Defensive Security

Blue Team defends against attacks through monitoring, detection, and response. Work is both reactive (incidents) and proactive (building defenses).

- **SIEM Expertise:** Log analysis, alert tuning, correlation rules
- **Incident Response:** NIST/SANS handling procedures
- **Threat Detection:** Attack patterns and anomaly identification
- **Networking:** Deep protocol and traffic analysis knowledge

Start Here: Set up ELK Stack → Generate sample logs → Write your first detection rule → Study NIST incident response framework.

Red Team - Offensive Security

Red Team attacks systems with authorization to find vulnerabilities before attackers do. Requires patience for deep technical learning.

- **Web Application Security:** OWASP Top 10, injection attacks
- **Exploitation:** Metasploit, manual exploitation techniques
- **Reporting:** Clear findings for non-technical stakeholders
- **Linux Internals:** System exploitation and privilege escalation

Start Here: Master networking/Linux → Study OWASP Top 10 → Practice on HackTheBox/WebGoat → Write a professional pentest report.

GRC - Governance, Risk & Compliance

Less technical but requires strong business acumen. Excellent entry-level option for career changers from non-IT backgrounds.

- **Frameworks:** NIST CSF, ISO 27001, PCI-DSS, HIPAA
- **Risk Management:** Identifying, assessing, and mitigating risks
- **Policy Development:** Creating security policies and procedures
- **Auditing:** Conducting and managing compliance audits

Start Here: Study NIST Cybersecurity Framework → Learn risk assessment → Build a risk assessment for a hypothetical organization.

Cloud Security

Rapidly growing as organizations migrate to cloud. High demand, excellent salaries, and genuine entry-level openings.

- **Cloud Platforms:** Pick one to start - AWS, Azure, or GCP
- **IAM:** Identity and access management in cloud environments
- **Network Security:** VPCs, security groups, network ACLs
- **Logging:** CloudTrail, CloudWatch, Azure Monitor

Start Here: Create free AWS account → Learn compute/storage/networking → Study IAM → Deploy a secure web application with proper logging.

AppSec - Application Security

Secures software development. Requires understanding of coding and secure development practices. Works closely with development teams.

- **Coding:** Python, Java, or JavaScript fundamentals
- **OWASP Top 10:** Common web application vulnerabilities
- **SAST/DAST:** Static and dynamic analysis tools
- **Secure SDLC:** Integrating security into development lifecycle

Start Here: Learn Python → Study OWASP Top 10 → Practice code review on open-source projects → Document findings professionally.

DevSecOps

Integrates security into CI/CD pipelines. Bridges development and security teams. Essential for modern software delivery.

- **CI/CD Pipelines:** Jenkins, GitLab CI, GitHub Actions
- **Container Security:** Docker and Kubernetes security
- **Infrastructure as Code:** Terraform, CloudFormation
- **Secrets Management:** Protecting API keys and credentials

Start Here: Learn GitHub Actions → Study Docker security → Learn Terraform → Build a CI/CD pipeline with automated security scanning.

06 - Building a Cybersecurity Home Lab

A home lab is your personal security playground - where you practice, experiment, and build projects without risk. **Every security professional has a home lab.** Building one is essential for learning and portfolio development.

Hardware Requirements

CPU

4+ cores - Intel i5/i7 or AMD Ryzen 5/7

RAM

16GB minimum - 32GB recommended for multiple VMs

Storage

512GB SSD minimum - 1TB+ for a full lab environment

Budget Option A

Use your existing computer. Start with 2-3 VMs. Upgrade RAM if possible.

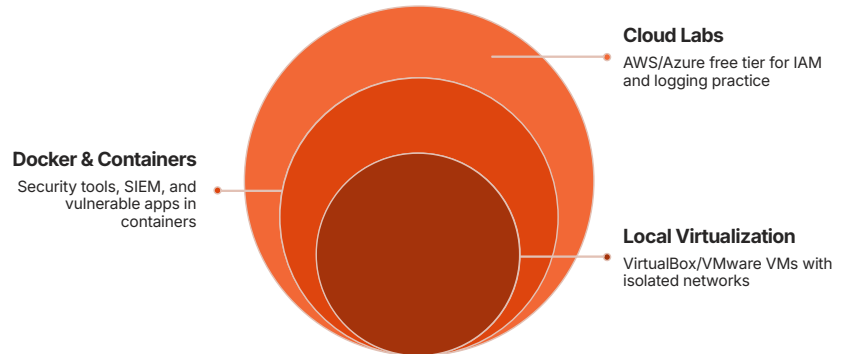
Budget Option B

Dedicated lab machine (\$800-\$1500). Used enterprise hardware is affordable.

Budget Option C

AWS/Azure free tier. Cheaper than hardware for small cloud-focused labs.

Three-Layer Lab Architecture



Start with Layer 1 and progress to Layers 2 and 3 as your skills develop. Each layer adds new tools, techniques, and project opportunities to your portfolio.

Three Flagship Lab Projects

1

Active Directory Lab

Objective: Set up a Windows domain with Active Directory.

1. Create a Windows Server VM as domain controller
2. Create Windows 10 client VMs and join to domain
3. Install and configure Active Directory, create users/groups/OUs
4. Configure Group Policy Objects (GPOs)

Skills Demonstrated: Windows administration, Active Directory, GPO, user management

2

SIEM Lab (ELK Stack)

Objective: Set up a SIEM and collect logs from multiple sources.

1. Deploy ELK Stack in Docker (Elasticsearch, Logstash, Kibana)
2. Configure log collection from Windows and Linux VMs
3. Create dashboards and visualizations in Kibana
4. Write detection rules and test with sample attack logs

Skills Demonstrated: SIEM configuration, log analysis, detection engineering

3

Vulnerable App Lab

Objective: Deploy vulnerable applications and practice finding vulnerabilities.

1. Deploy OWASP Juice Shop and WebGoat in Docker
2. Use Burp Suite to intercept and analyze HTTP traffic
3. Identify and exploit vulnerabilities in a controlled environment
4. Write a professional security report documenting findings

Skills Demonstrated: Web security, vulnerability assessment, security reporting



Documentation is your portfolio: Take screenshots, write notes, create network diagrams. Store everything on GitHub (without sensitive data) to showcase your work to employers.

07 - Projects, Portfolio & Specialization Project Ideas

Certifications and degrees open doors, but **projects close the deal**. Employers want to see what you can actually do. A strong portfolio demonstrates practical skills, problem-solving ability, and communication. Every project should tell a story: what problem you solved, how you solved it, and what you learned.

Project Ideas by Specialization Track

Track	Top 3 Project Ideas
Blue Team	SIEM detection rule (brute force pattern) · Incident response report from simulated breach · Network monitoring dashboard with anomaly alerting
Red Team	Penetration test report on Juice Shop/WebGoat · Vulnerability assessment with prioritized remediation plan · Metasploit exploitation lab with documented attack chain
Cloud Security	Secure web application deployment in AWS with IAM + logging · Cloud IAM design for hypothetical org · Cloud security audit documenting misconfigurations
AppSec	Security code review of open-source project · OWASP Top 10 analysis with remediation steps · API security assessment with documented findings
DevSecOps	Secure CI/CD pipeline with SAST + DAST + dependency scanning · Container vulnerability scanning documentation · Secrets management implementation in GitHub Actions
GRC	Risk assessment for hypothetical organization with NIST framework · Security policy for access control or incident response domain · Compliance audit against ISO 27001 controls

What Every Project Must Include

1 Objective

What problem are you solving? State this clearly in your README.

2 Methodology

How did you approach the problem? Tools used, steps taken.

3 Findings

What did you discover? Vulnerabilities, misconfigurations, risks.

4 Screenshots

Visual evidence of your work. Non-negotiable for credibility.

5 Lessons Learned

What was difficult? How did you overcome it? Shows growth mindset.

Portfolio Platforms and Quality Standards



GitHub

Create a repository for every project. Write detailed README files with objectives, methodology, findings, and screenshots. Keep code clean and well-commented. Make repositories public.



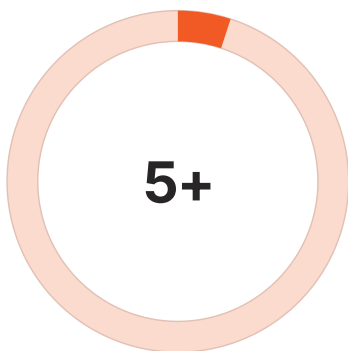
Personal Blog or Website

Write detailed project write-ups explaining your methodology and findings. Include diagrams and screenshots. A blog demonstrates communication skills - critical for security roles.



LinkedIn

Link to your GitHub and blog. Share project summaries. Engage with security content. Follow security professionals and companies. Your LinkedIn is your professional brand.



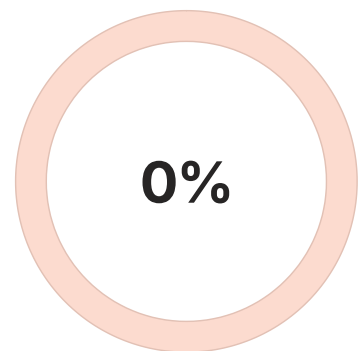
Min Projects

Complete at least 5 projects spanning your chosen specialization before applying.



Documented

Every project needs detailed documentation - no exceptions.

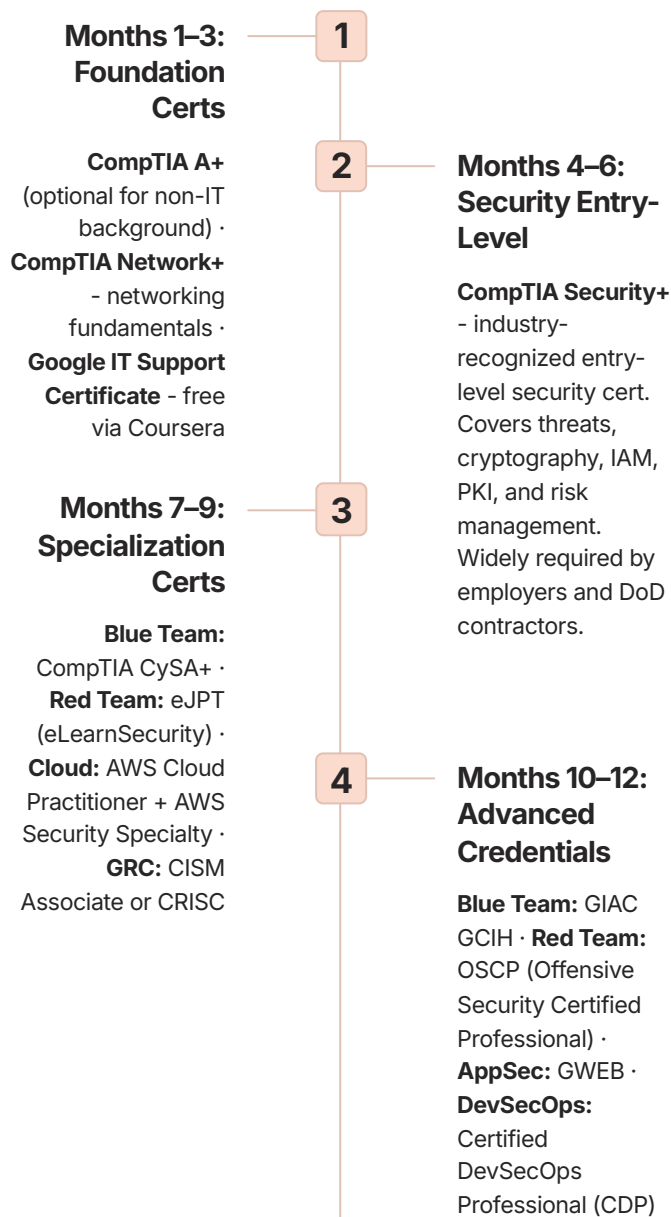


Plagiarism

All projects must be your own original work. Employers can tell the difference.

08 - Certifications, Learning Strategy & Job Search

Certification Roadmap by Track



Resume and LinkedIn Strategy

Your resume must be **one page**, **ATS-optimized**, and **project-focused**. Avoid generic summaries. Emphasize measurable achievements and specific tools.

Resume Structure

Header: Name, LinkedIn URL, GitHub URL, location (no street address)

Summary: 2–3 lines - target role, key skills, and strongest differentiator

Skills: Organized by category (SIEM, Networking, Cloud, Languages)

Projects: 3–5 projects with specific tools and outcomes

Certifications: List with dates

Education: Degrees and relevant bootcamps

LinkedIn Optimization

Headline: "Aspiring SOC Analyst | Security+, Network+ | ELK Stack, Nmap, Wireshark" - specific, keyword-rich

About: Career story, why security, what you build

Featured: Link to GitHub portfolio and best project write-up

Activity: Comment on security posts weekly. Share project updates. Follow SANS, CISA, threat intel accounts.

Job Search Channels

LinkedIn Jobs: Set alerts for "SOC Analyst", "Security Analyst", "Cloud Security"

Indeed / CyberSecJobs: Filter for entry-level and "0–2 years experience"

Networking: Join ISACA, (ISC)², OWASP local chapters. Attend DEF CON, BSides events.

Direct Outreach: Message security professionals on LinkedIn for informational interviews

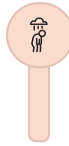
Free and Paid Learning Resources

Platform	Best For	Cost	Tracks
TryHackMe	Beginners, guided labs	Free + \$14/mo	Blue Team, Red Team
HackTheBox	Intermediate+, challenges	Free + \$14/mo	Red Team, AppSec
Coursera / edX	Structured courses, certs	Free audit + \$49/mo	All tracks
AWS Training	Cloud security	Free tier available	Cloud Security
SANS Free Resources	Threat intel, forensics	Free	Blue Team, GRC
Professor Messer	CompTIA exam prep	Free	All entry-level

09 - Interview Preparation

Technical knowledge gets you the interview. **Communication and preparation get you the offer.** Security interviews test both technical depth and your ability to explain complex concepts clearly. Prepare for three interview types: behavioral, technical, and scenario-based.

The STAR Framework for Behavioral Questions



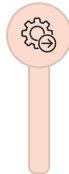
Situation

Set the context. Describe the challenge or scenario you faced. Keep it brief - 1-2 sentences.



Task

Describe your specific responsibility. What were you accountable for in this situation?



Action

Explain what YOU specifically did. Use "I" not "we." Be specific about tools and methods.



Result

Quantify the outcome. What improved? What was prevented? What did you learn?

STAR Example - "Tell me about a time you solved a difficult technical problem."

S: While building my home SIEM lab, my ELK Stack stopped ingesting logs after a configuration change.

T: I needed to diagnose and fix the issue without wiping my environment - I had 3 projects documented inside.

A: I reviewed Logstash pipeline logs, identified a parsing error in my grok pattern, corrected the filter, and tested with a small log sample before restarting the full pipeline.

R: Log ingestion resumed within 90 minutes. I documented the fix in my GitHub README so others could learn from it.

Common Interview Questions & Model Answers

"Explain the CIA Triad."

CIA stands for Confidentiality, Integrity, and Availability. Confidentiality ensures only authorized parties access data - enforced through encryption and access control. Integrity ensures data hasn't been tampered with - enforced through hashing and digital signatures. Availability ensures systems are accessible when needed - enforced through redundancy and disaster recovery. Together they form the foundation of every security decision.

"What is the difference between authentication and authorization?"

Authentication (AuthN) verifies *who you are* - through passwords, biometrics, or MFA. Authorization (AuthZ) determines *what you're allowed to do* - based on roles and permissions. For example, a user may authenticate successfully but still be unauthorized to access a sensitive file because their role doesn't have that permission.

"Walk me through how you would respond to a phishing alert."

First, I would triage the alert: confirm it's a true positive by examining email headers, sender domain, and links. Next, I would contain the threat: quarantine the email from the mail server and block the malicious domain at the firewall. Then I would assess impact: determine how many users received it and whether any clicked. Finally, I would remediate: reset credentials for affected users, document the incident, and write a detection rule to catch similar emails in the future.

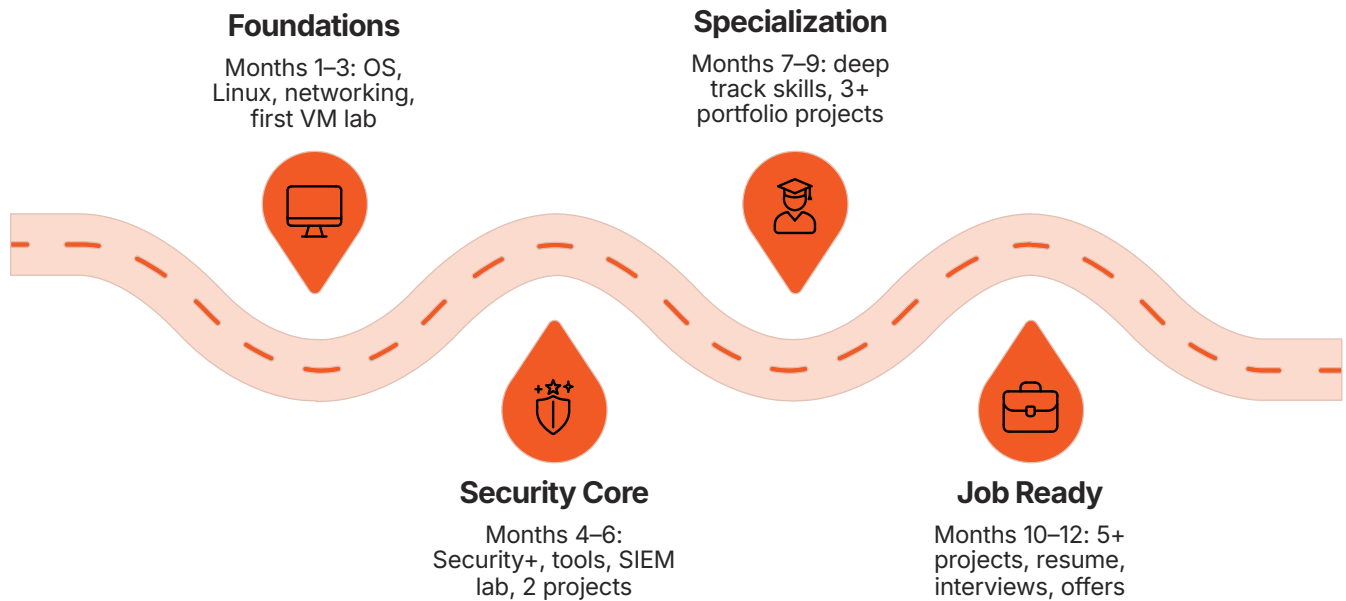
"What security tools are you comfortable with?"

Always tie your answer to hands-on experience: "I've used Nmap extensively for network scanning in my home lab, Wireshark for packet analysis, Burp Suite Community Edition for web application testing against OWASP Juice Shop, and ELK Stack for log ingestion and detection rule creation. I've documented all of these projects on my GitHub."

Technical Question Topics to Prepare

- OSI model - explain each layer and its security implications
- TCP handshake - three-way handshake and SYN flood attacks
- Common attack types - phishing, SQL injection, XSS, MITM, ransomware
- Defense in depth and Zero Trust architecture
- Log analysis - identify suspicious patterns in sample logs
- Encryption - symmetric vs asymmetric, TLS, certificate authorities

10 — 12-Month Roadmap & Career Readiness Checklist



Month-by-Month Milestones

Month	Focus	Deliverable / Milestone
1	Computer & OS fundamentals	Linux VM running, 15+ CLI commands mastered, GitHub account created
2	Networking (OSI, TCP/IP, DNS)	Networking+ content complete, Wireshark packet captures documented
3	Security fundamentals + CIA triad	Security+ study begins, first Nmap scanning project on GitHub
4	Security+ exam prep + tools	Security+ passed, Burp Suite + WebGoat lab complete
5	Specialization track selected	SIEM lab or equivalent specialization project started
6	First 2 portfolio projects complete	2 documented projects on GitHub, LinkedIn profile optimized
7	Specialization deep dive	Specialization cert study begins (CySA+, eJPT, AWS, etc.)
8	Projects 3–4 complete	4 documented projects, resume finalized, 5 job applications sent
9	Specialization cert achieved	Second cert earned, 20+ applications sent, networking active
10	Project 5 + Interview prep	5 projects complete, 10 mock interviews completed, STAR stories ready
11	Active interviewing	50+ applications, active phone screens and technical interviews
12	Offer & negotiation	First cybersecurity offer received and negotiated

Career Readiness Checklist

Before applying broadly, verify you have completed the following across all domains:

Foundations ✓

- ✓ Linux and Windows fundamentals mastered
- ✓ Networking — OSI, TCP/IP, DNS, ports
- ✓ CIA Triad, Zero Trust, Defense in Depth

Technical Skills ✓

- ✓ Nmap, Wireshark, Burp Suite hands-on experience
- ✓ SIEM configured and detection rules written
- ✓ Git and GitHub proficiency
- ☐ Specialization-specific tool mastery

Portfolio ✓

- ✓ 5+ completed, documented projects on GitHub
- ✓ All projects include screenshots and findings
- ☐ Personal blog or project write-ups published

Certifications ✓

- ✓ CompTIA Security+ achieved
- ☐ Specialization cert in progress or earned

Job Search ✓

- ✓ Resume (one page, ATS-optimized, project-focused)
- ✓ LinkedIn headline, About, and Featured sections complete
- ✓ STAR framework stories prepared (6–8 stories)
- ✓ Technical interview topics studied and practiced

Mindset ✓

- ☐ **Final Reminder:** Security is a journey, not a destination. Entry-level is the start, not the end goal. Stay curious, keep building, and contribute back to the community that helped you grow. Your career in cybersecurity starts the moment you take the first step.